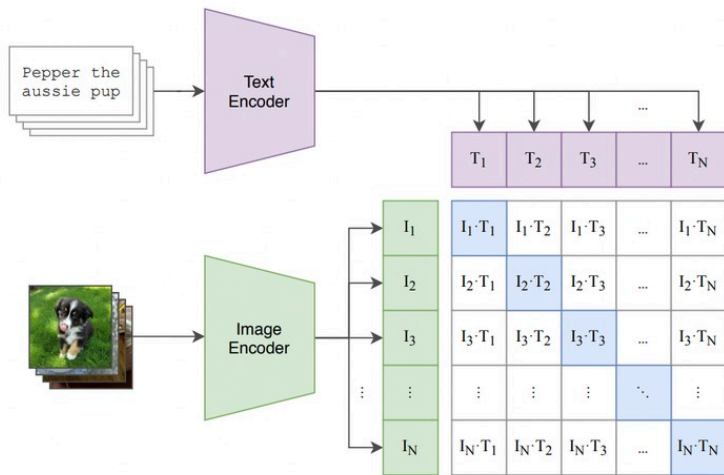
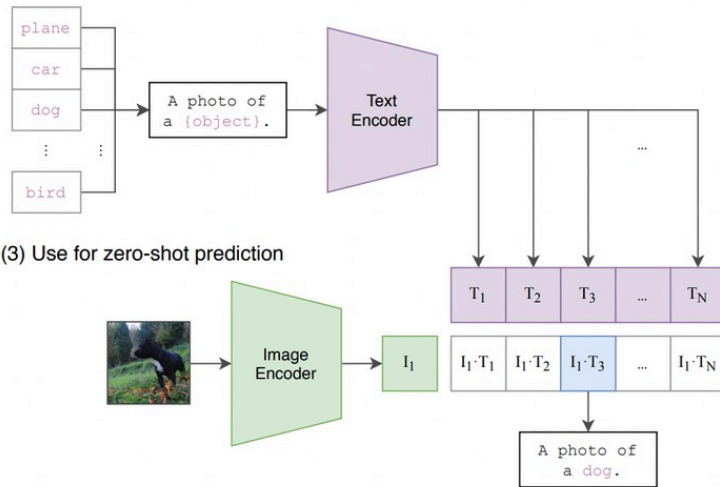


CLIP

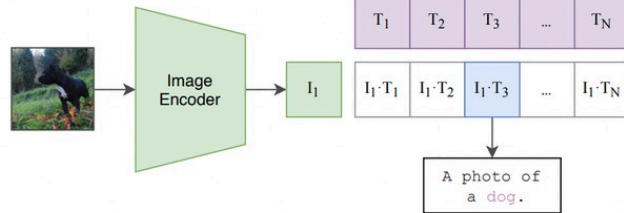
(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

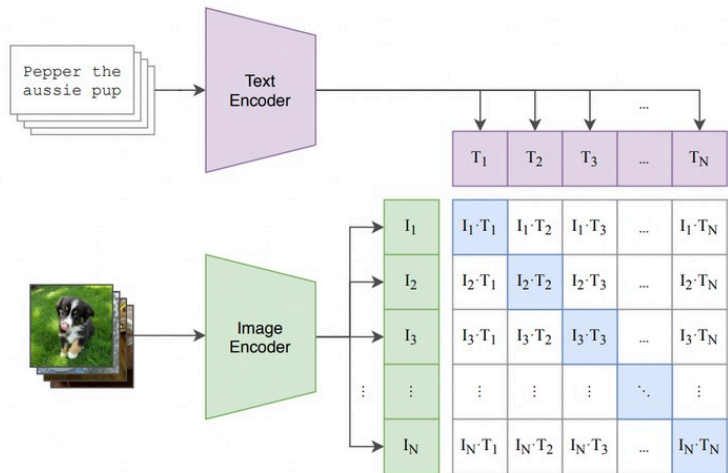


What, why and how?

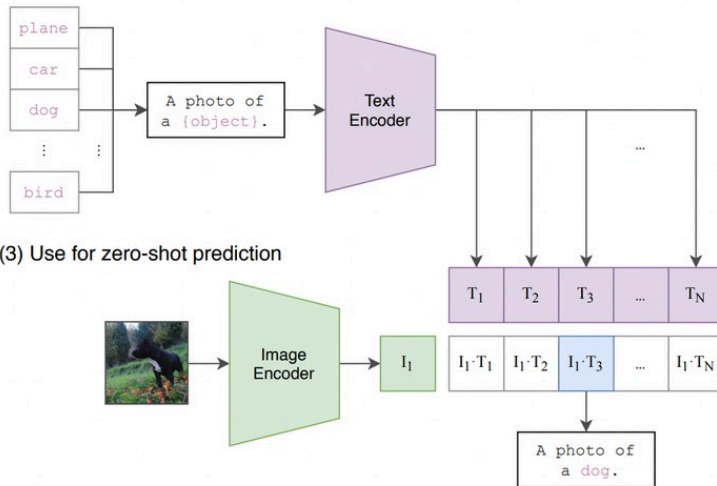
What is CLIP?

CLIP = Contrastive Language Image Pre-training

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

What is CLIP?

CLIP a neural network that jointly trains an image encoder and text encoder to map respective modalities to the same embedding space

What is CLIP?

CLIP a neural network that jointly trains an image encoder and text encoder to map respective modalities to the same embedding space

Let's understand how and why we do this...

Training CLIP

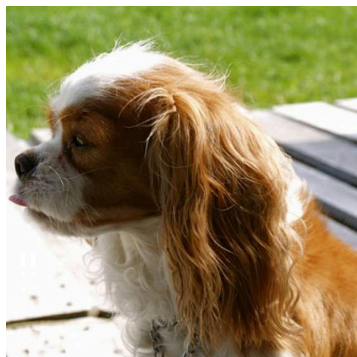
Consider a training set of image + free text pairs



This mural of a siberian huskey is so beautiful



The 2012 Honda accord has 160K miles on it.



My cute sally is the best dog in the world.



We got lucky seeing antelope fighting.

Training CLIP

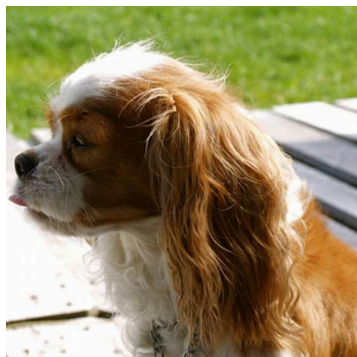
Create batches of size N .



This mural of a siberian huskey is so beautiful



The 2012 Honda accord has 160K miles on it.

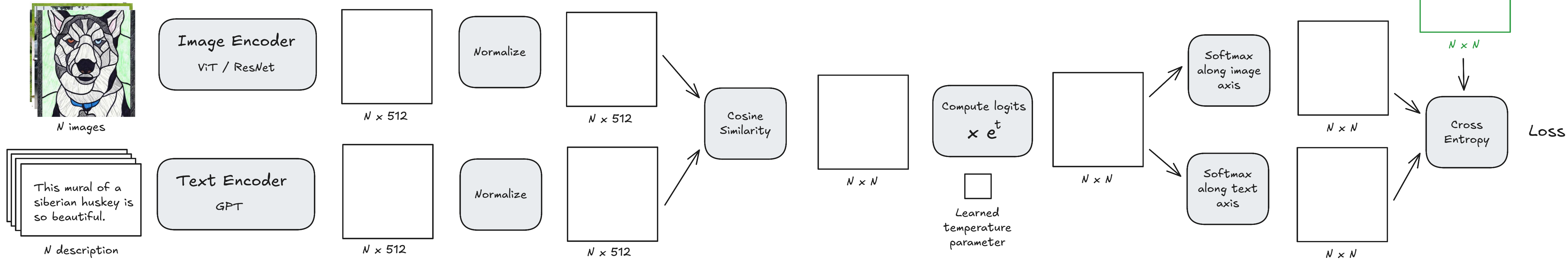


My cute sally is the best dog in the world.

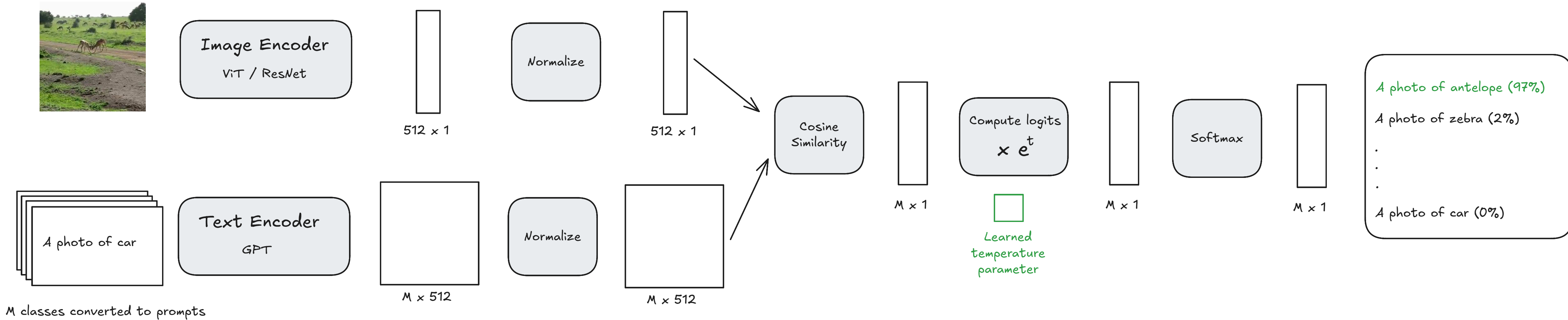


We got lucky seeing antelope fighting.

Training CLIP



Zero Shot Inference



Why CLIP?

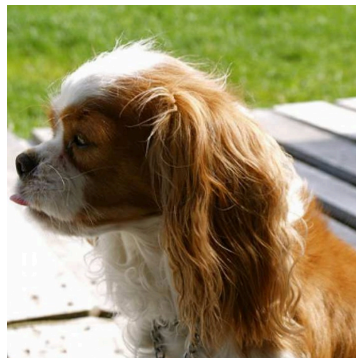
1. Task specific labels can be hard to come by



Dog



Car



Dog

Why CLIP?

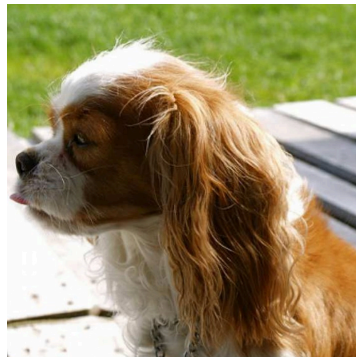
1. Task specific labels can be hard to come by



This mural of a siberian huskey is so beautiful



The 2012 Honda accord has 160K miles on it.



My cute sally is the best dog in the world.

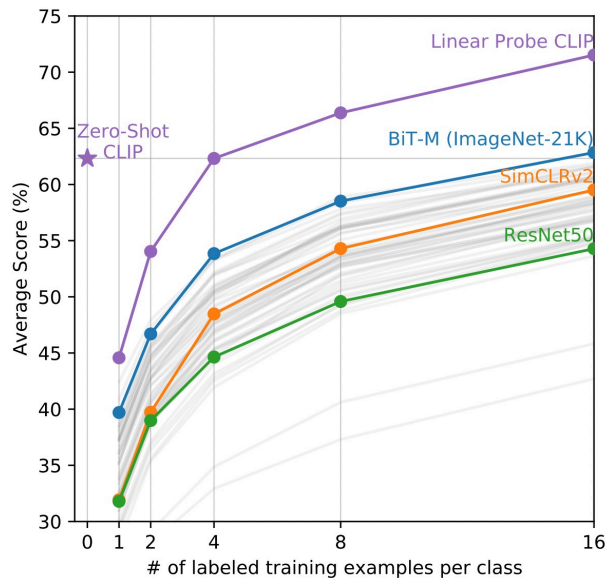
It is far easier to get natural language labels

Why CLIP?

2. Natural Language supervision allows the image encoder to create rich vectors that better encode the meaning of the image.

Let's take a look at some code to show how

Performance



During inference, Zero shot CLIP outperforms other networks (convolutional or transformer) that are trained with golden labels.

Linear Probing

Method to evaluate the visual representations of CLIP Image encoder. It involves training a linear model (probe) on top of the frozen CLIP encoder.

Linear Probing

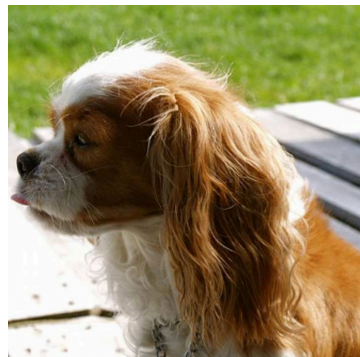
Say we have a dataset with classification labels



Dog



Car

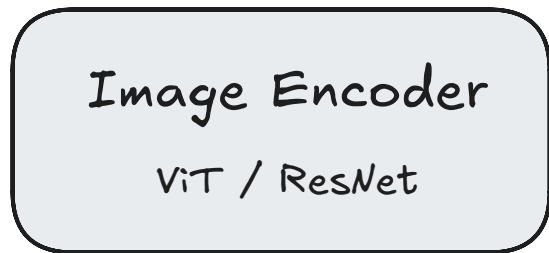


Dog

Say we have a 10 class classification task

Linear Probing

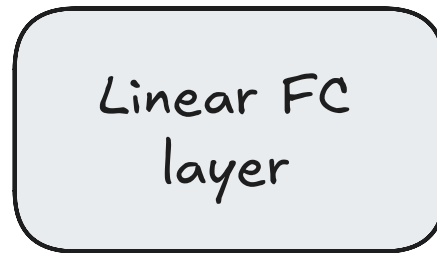
Add a linear feed forward layer and train



Trained



512×1



untrained
"probe"

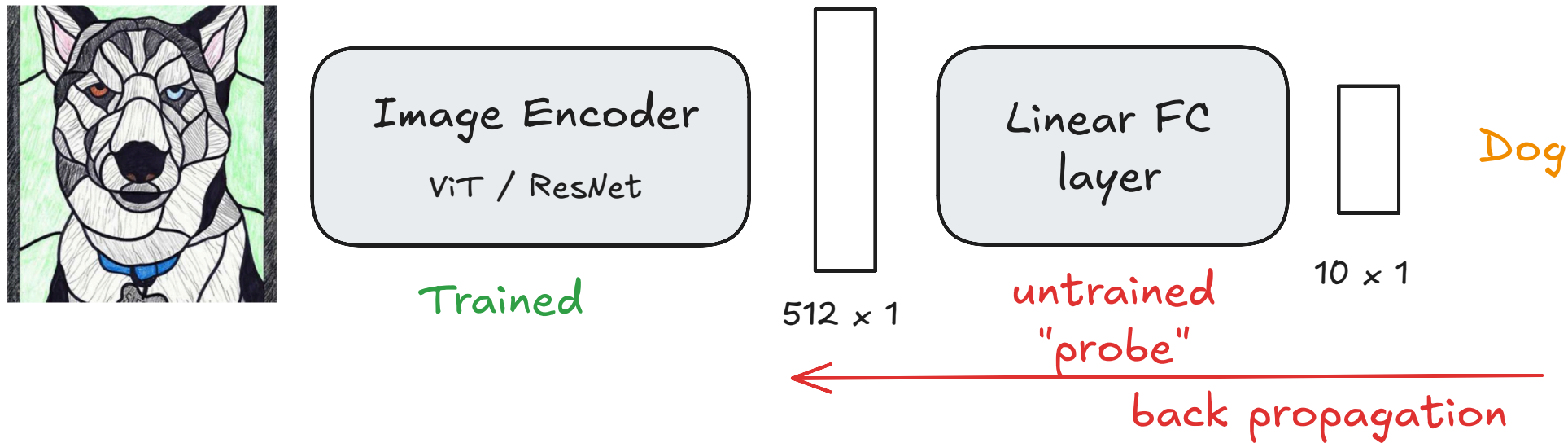


10×1

Dog

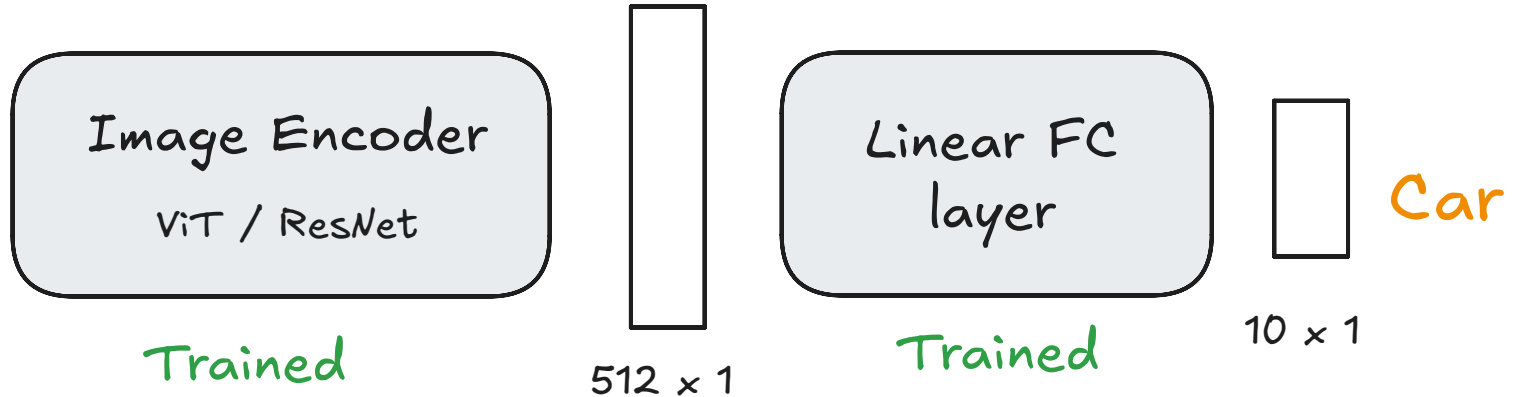
Linear Probing

Freeze CLIP encoder weights and update only the FC weights



Linear Probing

Once trained, the probe can help understand how well the CLIP encoder performs on the new dataset.



Linear Probing

It can even improve performance of CLIP Encoder on that specific dataset if data for each category is easy to come by.

